



Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

Q1. Read the following items carefully and encircle the correct option listed below at
(One mark)

- i. In SPSS, what is the 'Data Viewer'?
 - a) A table summarizing the frequencies of data for one variable.
 - b) A spreadsheet into which data can be entered.
 - c) A dialog box that allows you to choose a statistical test
 - d) screen in which variables can be defined and labelled.
- ii. How is a variable name different from a variable label?
 - a) It is shorter and less detailed.
 - b) It is abstract and unspecific.
 - c) It is longer and more detailed
 - d) It refers to codes rather than variables
- iii. What does the operation 'Recode Into Different Variables' do to the data?
 - a) Replaces missing data with some random scores.
 - b) Reverses the position of the independent and dependent variable on a graph.
 - c) Redistributes a range of values into a new set of categories and creates a new variable.
 - d) Represents the data in the form of a pie chart.
- iv. How would you use the drop-down menus in SPSS to generate a frequency table?
 - a) Open the Output Viewer and click: Save As → Pie Chart
 - b) Click on: Analyze → Descriptive Statistics → Frequencies.
 - c) Click on: Graphs → Frequencies → Pearson.
 - d) Open the Variable Viewer and recode the value labels.
- v. In which sub-dialog box can the Chi Square test be found?
 - a) Frequencies: Percentages.
 - b) Bivariate: Pearson.
 - c) Crosstabs: Statistics.
 - d) Gender: Female.
- vi. There are basic windows in SPSS.
 - a) 1
 - b) 2
 - c) 3
 - d) 4
- vii. In spss we can use types of variables.
 - a) 4
 - b) 6
 - c) 7
 - d) 8
- viii. menu allows you to open, print, and save data files and results, to close files and to quit SPSS.
 - a) Data
 - b) File
 - c) Analyze
 - d) Transform
- ix. The SPSS data editor provides number of spread sheets.
 - a) 2
 - b) 3
 - c) 4
 - d) None of the above
- x. The recode command is available in SPSS undermenu bar.
 - a) File
 - b) Analyze
 - c) Transform
 - d) Data



UNIVERSITY OF THE PUNJAB

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Roll No.

PAPER: Statistical Computer Packages
Course Code: STAT-309

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE SHORT QUESTIONS

Q2. Define the following:

(5 marks each)

- Scales of Measurement and their Types
- Rules of defining a variable name in SPSS
- Components of a Computer
- Types of Windows in SPSS

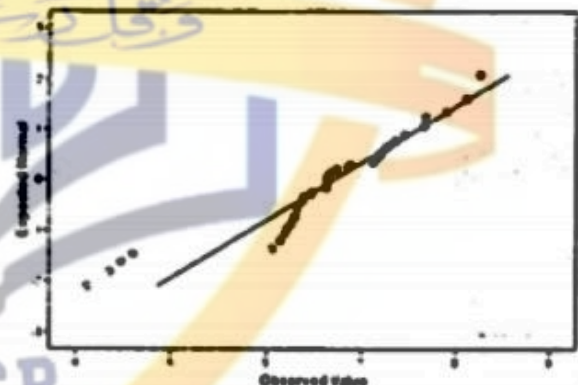
SUBJECTIVE

Note: You are required to justify your answers in this section by stating hypothesis, statistics, reference of graphs etc. on your answer sheet.

Q3. a) The following are some output generated through SPSS from a data set containing 56 observations.

	Statistic	Std. Error
Mean	6.7080	.11773
95% Confidence Interval for Mean		
Lower Bound	6.4721	
Upper Bound	6.9440	
5% Trimmed Mean	6.7801	
Median	6.6350	
Variance	.778	
Std. Deviation	.88102	
Minimum	4.12	
Maximum	8.26	
Range	4.14	
Interquartile Range	.99	
Skewness	-.901	.319
Kurtosis	1.710	.628

Normal Q-Q Plot of VAR00001



	Kolmogorov-Smirnov			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
VAR00001	.170	56	.000	.905	56	.000

VAR00001	
Chi-Square	14.000
df	39
Asymp. Sig.	1.000

	t	df	Sig. (2-tailed)	95% Confidence Interval of the Difference	
				Lower	Upper
VAR00001	-2.480	55	.016	-.5279	-.0560

VAR00001	
Test Value(a)	6.64
Cases < Test Value	28
Cases >= Test Value	28
Total Cases	56
Number of Runs	21
Z	-2.158
Asymp. Sig. (2-tailed)	.031

Use the above output and answer the following questions.

- Does the data used in analysis follows a normal distribution?
- Can we say that the data is randomly distributed?
- Does the data significantly differing from 7.0?

- b) Three sections of the same elementary mathematics course are taught by three teachers A, B and C. The final grades were recorded and the following output is generated through SPSS:

Grades			
Levene Statistic	df1	df2	Sig.
1.956	2	42	.154

Grades	
Chi-Square	.601
df	2
Asymp. Sig.	.740

a. Kruskal Wallis Test

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	142.178	2	71.089	.217	.805
Within Groups	13728.133	42	326.860		
Total	13870.311	44			

- Test the Homogeneity of Grades among three Teachers.
- Can we conclude that there is no significant difference among the Grades of three Teachers? (2+2)

Q4. a) Sixty candidates for scientific posts gave particulars of their university degrees and their hobbies. The degrees were in mathematics, statistics or economics, and the hobbies could be classified roughly as music, craftwork, reading or drama. The following output is generated from the recorded data using SPSS:

			Mathematics	Statistics	Economics	Total
Hobbies	Music	Count	5	2	3	10
		Expected Count	3.5	3.2	3.3	10.0
		% within Hobbies	50.0%	20.0%	30.0%	100.0%
		% within Subject	23.8%	10.5%	15.0%	16.7%
		% of Total	8.3%	3.3%	5.0%	16.7%
	Craftwork	Count	6	6	7	21
		Expected Count	7.4	6.7	7.0	21.0
		% within Hobbies	28.6%	38.1%	33.3%	100.0%
		% within Subject	28.6%	42.1%	35.0%	35.0%
		% of Total	10.0%	13.3%	11.7%	35.0%
	Book Reading	Count	4	6	5	15
		Expected Count	5.3	4.8	5.0	15.0
		% within Hobbies	26.7%	40.0%	33.3%	100.0%
		% within Subject	19.0%	31.6%	25.0%	25.0%
		% of Total	6.7%	10.0%	8.3%	25.0%
	Drama	Count	6	3	5	14
		Expected Count	4.9	4.4	4.7	14.0
		% within Hobbies	42.9%	21.4%	35.7%	100.0%
		% within Subject	28.6%	15.8%	25.0%	23.3%
		% of Total	10.0%	5.0%	8.3%	23.3%
Total		Count	21	19	20	60
		Expected Count	21.0	19.0	20.0	60.0
		% within Hobbies	35.0%	31.7%	33.3%	100.0%
		% within Subject	100.0%	100.0%	100.0%	100.0%
		% of Total	35.0%	31.7%	33.3%	100.0%

		Value	Approx. Sig.
Nominal by Nominal	Contingency Coefficient	.218	.810
N of Valid Cases		60	

- What is the percentage of statistics students having craftwork as a hobby from all statisticians?
- What percentage of Math students like book reading from all students like book reading?
- What is the residual value of Economics students with Drama as a hobby?
- Can we conclude that selection of hobbies does not depend upon the subject? (1+1+1+3)

- b) A study was conducted to examine the effect of blood group on individual's weight. Two groups of individuals with +ve and -ve blood group were selected randomly from a large population and their weights (kg) were recorded. Some output generated from the recorded data is given below:

	Group	N	Mean	Std. Deviation	Std. Error Mean
Weight	+ve	20	37.3000	8.44861	1.88917
	-ve	20	39.9000	12.11309	2.70856

		Levene Test for Equality of Variances		t-test for Equality of Means						95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference		Lower	Upper
Weight	Equal variances assumed	3.248	.080	-.787	38	.438	-2.60000	3.30231		-8.28518	4.08518
	Equal variances not assumed			-.787	33.948	.437	-2.60000	3.30231		-8.31148	4.11148

	Weight
Mann-Whitney U	180.000
Wilcoxon W	390.000
Z	-.542
Asymp. Sig. (2-tailed)	.588
Exact Sig. [2*(1-tailed Sig.)]	.602(a)

	Group - Weight
Z	-.691(a)
Asymp. Sig. (2-tailed)	.490

a. Based on negative ranks.
b. Wilcoxon Signed Ranks Test

		Paired Differences			95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			
Pair 1	Weight - Group	-2.60000	14.53634	3.25043	-8.40322	4.20322	-.808	18	.434

- Test the Homogeneity of Variance between the two blood groups.
- Can we conclude that there is no significant difference in the average weight between +ve blood group and -ve blood group persons? (2+2)

Q5. An investigator believes that individual's age (X_1), income (X_2) and savings (X_3) are the major factors of insurance policies purchase (Y). The following output is generated from the recorded data using SPSS:

		Y	X1	X2	X3
Y	Pearson Correlation	1	.892	.908	.925
	Sig. (2-tailed)		.000	.000	.000
	N	27	27	27	27
X1	Pearson Correlation	.89	1	.990	.988
	Sig. (2-tailed)	.000		.000	.000
	N	27	27	27	27
X2	Pearson Correlation	.908	.990	1	.993
	Sig. (2-tailed)	.000	.000		.000
	N	27	27	27	27
X3	Pearson Correlation	.925	.988	.993	1
	Sig. (2-tailed)	.000	.000	.000	
	N	27	27	27	27

Control Variables			Y	X1
X2 & X3	Y	Correlation	1.000	-.313
		Significance (2-tailed)		.128
		df	0	23
	X1	Correlation	-.313	1.000
		Significance (2-tailed)	.128	
		df	23	0

Control Variables*			Y	X3
X2 & X1	Y	Correlation	1.000	.552
		Significance (2-tailed)		.004
		df	0	23
	X3	Correlation	.552	1.000
		Significance (2-tailed)	.004	
		df	23	0

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.938(a)	.877	.861	15.51364	.763

a Predictors: (Constant), X3, X1, X2

b Dependent Variable: Y

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	39349.189	3	13116.396	54.499	.000(a)
	Residual	5535.477	23	240.673		
	Total	44884.667	26			

a Predictors: (Constant), X3, X1, X2

b Dependent Variable: Y

Coefficients(a)

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
							Tolerance	VIF
1	(Constant)	387.370	44.192		8.766	.000		
	X1	-4.496	2.840	-.859	-1.576	.128	.018	55.259
	X2	-1.211	3.096	-.264	-.391	.699	.012	85.076
	X3	38.664	12.179	2.036	3.175	.004	.013	76.687

a Dependent Variable: Y

- Is the correlation between Income and Policies Purchase significant?
- Is the correlation between savings and Policies Purchase significant when age and income is controlled?
- Fit a regression model $Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + e$ and interpret the results. Also test the significance of the regression coefficients and the intercept term.
- Test the significance of the regression model given in (iii). (2+2+4+2)